$\mathsf{D}A$ as a limit of the $\mathsf{D}_N A$

strict limit, homotopy limit, and a tower without fibrations

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Abstract

For a type A of the poset model, $\mathsf{D}_N A$ is the localisation $\mathsf{D}A$ with the families indexed only by the simplices of dimension at most N. We show three facts. $\mathsf{D}A$ is the strict limit of the tower $(\mathsf{D}_N A)_N$. If A is fibrant, $\mathsf{D}A$ is also its homotopy limit: the comparison map from $\mathsf{D}A$ to the internal limit, a type built with Π, Σ and Path, is a homotopy equivalence. Finally, the maps $\mathsf{D}_{N+1}A \to \mathsf{D}_N A$ need not be fibrations, even when A is fibrant, so the second fact is not a consequence of the first. All three are checked in `core`.

1 Setting

The poset model. $\mathcal{C}$ is the category of finite non-empty posets and monotone maps, and $\Delta \subseteq \mathcal{C}$ the full subcategory of the linear posets $[n] = \{0 < \cdots < n\}$. A map of Δ is *strict* if it is strictly monotone, and *surjective* if it is onto. Every map g of Δ factors uniquely as $g = c\sigma$ with c strict and σ surjective (Eilenberg–Zilber), and the same holds for a map $g : [k] \to P$ into a poset, with $c : [r] \to P$ injective. A context Γ is a presheaf on $\mathcal{C}$, and a type A over Γ a presheaf on its category of elements; we write $A(X, \rho)$ for the fibre at $\rho \in \Gamma(X)$ and $a\,g$ for restriction. Every poset is identified with its nerve, so that $[m] \times [1]^k$ is $\Delta[m] \times \Delta[1]^k$. We write i^*A for the restriction of A to Δ, a simplicial set over $i^*\Gamma$, and $\partial[m]$ for the union of the faces of $[m]$.

A *path* in A at (X, ρ) from a to b is an element of A at $X \times [1]$ over $\rho\pi_1$ whose restrictions to $X \times \{0\}$ and $X \times \{1\}$ are a and b. A family of paths natural in X is a section of the type Path_A. A *horn filling operation* on a simplicial set T chooses fillers of horns $\Lambda^k[n] \to T$ such that the filler of the restriction of a horn along a face map is the restriction of the filler, whenever both are horns. We use two facts about fibrant types of the poset model.

(F1) If A is fibrant, i^*A has a horn filling operation.

(F2) If i^*B has a horn filling operation, the type $\mathsf{D}B$ defined below is fibrant, naturally in Γ.

Definition 1.1. Let A be a type over Γ. An element of $\mathsf{D}A$ at (X, ρ) is a family

$$u(f) \in A([k], \rho f), \qquad f : [k] \to X,$$

with $u(f)\,g = u(fg)$ for every strict $g : [j] \to [k]$. Restriction along $X' \to X$ is precomposition. For $N \geq -1$, an element of $\mathsf{D}_N A$ at (X, ρ) is the same with only the maps $f : [k] \to X$ with $k \leq N$. The map $r : \mathsf{D}_{N+1}A \to \mathsf{D}_N A$ forgets the values in dimension $N + 1$, and $v_N : \mathsf{D}A \to \mathsf{D}_N A$ forgets the values above N.

The important point is that the elements of $\mathsf{D}A$ and of $\mathsf{D}_N A$ are natural along the strict maps only. Their values on degenerate simplices are not determined by the values on the non-degenerate ones.

The coskeleton. An element of $K_N A$ at (X, ρ) is a family $c(f) \in A([k], \rho f)$, for $k \leq N$ and $f : [k] \to X$, with $c(f)\, g = c(fg)$ for every $g : [j] \to [k]$ with $j \leq N$. So $i^* K_N A$ is the N-coskeleton of $i^* A$, and $\eta(a)(f) = af$ defines $\eta : A \to K_N A$.

Lemma 1.2.

1. *For $m \leq N$, $\eta : A([m], \tau) \to K_N A([m], \tau)$ is a bijection, with inverse $c \mapsto c(\mathrm{id})$.*
2. *If $i^* A$ has a horn filling operation, so has $i^* K_N A$.*

Proof. (1) The identity of $[m]$ is one of the indices, and $c(g) = c(\mathrm{id})\, g$.

(2) Let h be a horn of $i^* K_N A$ on $\Lambda^k[m+1]$. If $m \leq N$, its faces are, by (1), elements of $A([m])$ forming a horn of $i^* A$; its filler a gives the filler $\eta(a)$. If $m > N$, a map $[j] \to [m+1]$ with $j \leq N$ misses at least two vertices, hence a vertex other than k, so it lies in the horn, and the filler is h read at these maps. Both cases commute with restriction to faces. □

Proposition 1.3. $\mathsf{D}_N A = \mathsf{D}(K_N A)$. *In particular $\mathsf{D}_N A$ is fibrant when A is.*

Proof. Given $u \in \mathsf{D}(K_N A)(X, \rho)$, put $w(f) = u(f)(\mathrm{id})$ for $f : [k] \to X$, $k \leq N$; it is natural along strict maps. Conversely, given $w \in \mathsf{D}_N A(X, \rho)$, put $u(f)(g) = w(fc)\, \sigma$, where $g = c\sigma$ is the factorisation of $g : [j] \to [k]$ with c strict and σ surjective. Two such factorisations are related by a strict map, which gives the naturality of $u(f)$ along every map and of u along strict maps. Below N the value $w(fc)\sigma$ is $w(f)\, c\sigma = w(f)g$, so the two maps are inverse. The second statement follows from (F1), Lemma 1.2(2) and (F2). □

2 The strict limit

Proposition 2.1. $\mathsf{D}A$ *is the limit of the tower $\cdots \to \mathsf{D}_{N+1} A \xrightarrow{r} \mathsf{D}_N A \to \cdots \to \mathsf{D}_{-1} A = 1$, with the cone $(v_N)_N$.*

Proof. A compatible sequence $(u_N)_N$, $u_N \in \mathsf{D}_N A(X, \rho)$, defines the family $u(f) = u_k(f)$ for $f : [k] \to X$. Compatibility gives $u_k(f) = u_N(f)$ for $k \leq N$, so u is natural along every strict map, since u_N is for N large. This u is the unique element of $\mathsf{D}A(X, \rho)$ with $v_N(u) = u_N$ for all N, and the construction commutes with restriction. □

3 The maps of the tower are not fibrations

Proposition 3.1. *There is a fibrant type A such that $r : \mathsf{D}_1 A \to \mathsf{D}_0 A$ has no fibration structure.*

Proof. Let A be the constant type with value $\{0, 1\}$ over the empty context, which is fibrant. Then $A([k]) = \{0, 1\}$ for every k, and every restriction is the identity. An element of $\mathsf{D}_0 A$ at X is an arbitrary function from the points of X to $\{0, 1\}$, since the only strict map $[0] \to [0]$ is the identity. An element of $\mathsf{D}_1 A$ at X is such a function together with a value on each map $[1] \to X$. Its value on an edge e restricts, along the two strict maps $[0] \to [1]$, to its values on the two ends of e. So the ends of every edge get the same value.

Take $X = [0]$. The element $w = (0)$ of $\mathsf{D}_1 A([0])$ lies over $a = 0$. The element ℓ of $\mathsf{D}_0 A([1])$ with $\ell(0) = 0$ and $\ell(1) = 1$ is a line of $\mathsf{D}_0 A$ from $r(w)$ to 1. A lift of ℓ starting at w is an element of $\mathsf{D}_1 A([1])$ over ℓ. It must give a value to the identity of $[1]$, an edge from 0 to 1, and there is none. So r does not even have the path lifting property. A fibration structure gives fillers for all open boxes over $[0] \times [1]$, and in particular such a lift. □

Remark 3.2. The same argument applies to $r : \mathsf{D}_{N+1}A \to \mathsf{D}_N A$ for every $N \geq 0$, for a fibrant A whose restriction to Δ has an N-sphere with no filler. Lifting a line of $\mathsf{D}_N A$ over $[N]$ asks for $(N+1)$-simplices of i^*A with a prescribed boundary. This is the coskeletal tower of a Kan complex: it converges strictly although its maps are not Kan fibrations.

So the strict limit of Proposition 2.1 is not the limit of a tower of fibrations, and it need not compute the homotopy limit. This is the content of the next section.

4 The homotopy limit

The *internal limit* of the tower is the type

$$L = \Sigma\,(u : \textstyle\prod_N \mathsf{D}_N A)\ \prod_N \mathrm{Path}_{\mathsf{D}_N A}(r\,u_{N+1}, u_N),$$

where the products range over the constant type of natural numbers. The *comparison* $\varphi : \mathsf{D}A \to L$ sends u to $((v_N u)_N, \text{the constant paths})$.

If A is fibrant, so is L: each $\mathsf{D}_N A$ is fibrant (Proposition 1.3), and fibrant types are closed under Path, Σ and products over the constant type of natural numbers, the latter componentwise.

Theorem 4.1. *If A is fibrant, φ is a homotopy equivalence: there is a map $\psi : L \to \mathsf{D}A$, natural in the context, with paths $\varphi\psi \simeq \mathrm{id}_L$ and $\psi\varphi \simeq \mathrm{id}_{\mathsf{D}A}$ in the model.*

The rest of this section proves the theorem. Write $C = i^*A$ and $C_N = i^*K_N A$, the N-coskeleton of C. We identify a poset with its nerve, so that $[m] \times [1]^k$ is $\Delta[m] \times \Delta[1]^k$.

4.1 Tools

Lemma 4.2 (Boxes)**.** *Let T be a simplicial set with a horn filling operation, P a finite poset, $S \subseteq P$ a simplicial subset and $e \in \{0,1\}$. Every map $P \times \{e\} \cup S \times \Delta[1] \to T$ extends to $P \times \Delta[1]$, with a choice of extension compatible with restriction along injective maps $P' \to P$.*

Proof. Prisms over a simplex. A problem at n is a map $b : [n] \times \{e\} \cup Q \times \Delta[1] \to T$ with $Q \subseteq [n]$; its restriction along d_i is a problem at $n-1$. By recursion on n we choose a filler $F(b)$ with $F(b)\,d_i = F(b\,d_i)$. If Q contains the identity, $F(b) = b$. Otherwise the fillers $F(b\,d_i)$ agree on common faces, and $\partial[n] \times \Delta[1] \cup [n] \times \{e\}$ is given. The non-degenerate $(n+1)$-simplices of $[n] \times [1]$ are the shuffles $P_k = ((0,0), \ldots, (k,0), (k,1), \ldots, (n,1))$, $0 \leq k \leq n$. The face d_k of P_k is the face d_k of P_{k-1}, the face d_{k+1} of P_k is the face d_{k+1} of P_{k+1}, and the other faces lie in $\partial[n] \times \Delta[1]$, apart from $d_{n+1}P_n = [n] \times \{0\}$ and $d_0 P_0 = [n] \times \{1\}$. For $e = 0$ fill $P_n, \ldots, P_0$ in this order, each as the filler of a horn on $\Lambda^k[n+1]$; for $e = 1$ fill $P_0, \ldots, P_n$, each on $\Lambda^{k+1}[n+1]$.

Prisms over a poset. For an injective $c : [n] \to P$ let b_c be the problem read along c, with $Q = c^*S$. Then $b_c\,d_i = b_{cd_i}$, so $F(b_c)\,\mu = F(b_{c\mu})$ for every injective μ. A simplex $g : [W] \to P$ factors uniquely as $g = c\sigma$, and we put $F(g,j) = F(b_c)(\sigma, j)$. If $k : [W'] \to [W]$, the factorisation of gk is $(c\mu)\sigma'$ with μ injective and $\mu\sigma' = \sigma k$, so $F(gk, jk) = F(g,j)\,k$. If $g \in S$ then $c \in S$, the identity is in c^*S, and $F(g,j)$ is the given value; the same holds at $j = e$. □

We apply the lemma with $P = [m] \times Y$ and $S = \partial[m] \times Y \cup [m] \times Q$, for a poset Y and a simplicial subset $Q \subseteq Y$. The *lid* is the value at e, the *sides* are the values on $\partial[m] \times Y$, and the *end data* are the values on $[m] \times Q$. For the squares, $Y = [1]_t$ and $Q = \{0,1\}$. For the cubes, $Y = [1]_t \times [1]_v$ and Q is the boundary of the square: four pieces at $t = 0$, $t = 1$, $v = 0$, $v = 1$, which must agree at the four corners. The fillers are uniform along the face maps, so no excluded middle is used.

Readings. Let P be a finite poset and $\phi : P \to X$. An element z of $\mathsf{D}A$ (or of $\mathsf{D}_N A$, if the simplices of P have dimension at most N) at X is *read* along ϕ as the map $P \to C$ given on an injective chain $c : [r] \to P$ by $z(\phi c)$, and extended to all simplices by the Eilenberg–Zilber factorisation $g = c\sigma$, with c injective and σ surjective. The reading is natural along all maps of Δ, although z is natural along strict maps only.

Let $u \in \mathsf{D}A(X)$ and $f : [m] \to X$, and put $U_f = u(f) \in C_m$. Reading u along $f\pi_1 : [m] \times [k] \to X$ gives a map $R_f^{[k]}$. For $k = 0$ it is the simplex U_f. We write $R_f = R_f^{[1]}$, a prism with both ends U_f.

Lemma 4.3. *R_f need not be the constant prism at U_f. There is a square Q_f over $[m] \times [1]_s \times [1]_v$ with*

$$Q_f(s,0) = R_f(s), \qquad Q_f(1,v) = R_f(v), \qquad Q_f(0,v) = R_f(v), \qquad Q_f(s,1) = U_f.$$

Readings commute with the restriction to faces: R_f, Q_f and U_f restricted along $d_i \times \mathrm{id}$ are R_{fd_i}, Q_{fd_i} and U_{fd_i}.

Proof. On the edge $(i,0) \to (i,1)$ of $[m] \times [1]$, R_f is u at a degenerate simplex of X. The naturality of u along strict maps says nothing about this value, which is not determined by $u(f)$.

Let $\beta : [1] \times [1] \to [2]$ be the monotone map with $\beta(s,0) = s$, $\beta(1,v) = 1 + v$, $\beta(0,v) = 2v$ and $\beta(s,1) = 2$. Put $Q_f = R_f^{[2]} \circ (\mathrm{id} \times \beta)$. The edges $v = 0$, $s = 1$ and $s = 0$ of β are the three strict maps $[1] \to [2]$. Reading along a strict inclusion $P' \to P$ is reading on P', since the injective chains of P' are injective chains of P. So these three edges of Q_f are R_f. The edge $v = 1$ is $R_f^{[2]}$ precomposed with $[m] \to [m] \times \{2\}$, which is again a strict inclusion, so it is the reading of u along f, which is U_f. Note that we precompose an already extended map. We do not claim that reading commutes with projections.

The last statement is the same argument for the strict inclusions $d_i \times \mathrm{id}$, together with $u(f)\, d_i = u(fd_i)$. □

4.2 The map ψ

Let $\ell = (u_N, p_N) \in L(X)$, so p_N is an element of $\mathsf{D}_N A$ at $X \times [1]$ from $r\, u_{N+1}$ to u_N. For $f : [m] \to X$, let $\sigma_N(f)$ be the reading of u_N along f and $p_N(f)$ the reading of p_N along $f \times [1]$. We build, by recursion on m and, for each f, on the stage $N \geq m + 1$:

- an m-simplex $\psi(f)$ of C;
- prisms $H^N(f)$ over $[m] \times [1]_t$ from $\psi(f)$ $(t = 0)$ to $\sigma_N(f)$ $(t = 1)$;
- squares $\Theta_N(f)$ over $[m] \times [1]_t \times [1]_s$ with $\Theta_N(t,1) = H^N(t)$, $\Theta_N(t,0) = H^{N+1}(t)$, $\Theta_N(0,s) = \psi(f)$ and $\Theta_N(1,s) = p_N(f)(s)$.

Each of these restricts along d_i to the same datum at fd_i.

1. $H^{m+1}(f)$ fills the box over $[m] \times [1]_t$ with lid $\sigma_{m+1}(f)$ at $t = 1$ and sides $H^{m+1}(fd_i)$. Its end at $t = 0$ is $\psi(f)$.
2. $\Theta_N(f)$ fills the box over $[m] \times [1]_t$, line s, with lid $H^N(f)$ at $s = 1$, end data $\psi(f)$ at $t = 0$ and $p_N(f)$ at $t = 1$, and sides $\Theta_N(fd_i)$. Its end at $s = 0$ is $H^{N+1}(f)$.

The face fd_i has dimension $m - 1$, so its data exist for $N \geq m$, and in particular at the stages $N \geq m + 1$ used at f. All these boxes lie in dimension at most $m + 2 \leq N + 1$. They live in C: the stage only says which of u_N and p_N is read. Then $\psi(\ell) = (f \mapsto \psi(f))$ is natural along the face maps, hence along all strict maps, and is an element of $\mathsf{D}A(X)$.

The value of $\psi(\ell)$ in dimension m is taken at stage $m+1$, never lower. In particular u_0 is never used, so the obstruction of Section 3 does not arise. The squares Θ_N are part of the data and are built by box filling, not by concatenating the paths p_N: a concatenation would not commute strictly with the faces.

4.3 The path $\varphi\psi \simeq \mathrm{id}_L$

A path in L from $\varphi\psi(\ell)$ to ℓ consists of lines l_N of $\mathsf{D}_N A$ from $v_N\psi(\ell)$ to u_N, and squares relating $r\, l_{N+1}$, l_N, the constant line and p_N. The line l_N is read off $H^N(f)$ at the maps $(f,h) : [k] \to X\times[1]$, $k \le N$, through $\langle \mathrm{id}, h\rangle$: its value at (f,h) is $H^N(f)$ at (id, h), natural along strict maps because H^N is face compatible. For $k < N$ the prism $H^N(f)$ exists. For $k = N$ it is built in C_N: $H^N(f)$ and $\Theta_N(f)$ fill, in C_N, the box formed by $H^{N+1}(f)$, $p_N(f)$, the constant line, and the sides $\Theta_N(fd_i)$. The squares are read off Θ_N in the same way, and their edges are the required ones.

4.4 The path $\psi\varphi \simeq \mathrm{id}_{\mathsf{D}A}$

Let $u \in \mathsf{D}A(X)$ and run the construction on $\varphi(u)$. Its path p_N is constant, so its reading $p_N(f)$ is R_f (Lemma 4.3). Thus $H^N(f)$ goes from $\psi\varphi(u)(f)$ to U_f, and

$$\Theta_N(t,1) = H^N(t), \quad \Theta_N(t,0) = H^{N+1}(t), \quad \Theta_N(0,s) = \psi\varphi(u)(f), \quad \Theta_N(1,s) = R_f(s).$$

We build, by recursion on m and on $N \ge m+1$, a prism $G(f)$ over $[m]\times[1]_t$ from $\psi\varphi(u)(f)$ to U_f and squares $\Xi^N(f)$ over $[m]\times[1]_t\times[1]_v$ with

$$\Xi^N(t,0) = H^N(t), \quad \Xi^N(t,1) = G(t), \quad \Xi^N(0,v) = \psi\varphi(u)(f), \quad \Xi^N(1,v) = R_f(v),$$

all restricting along d_i to the data at fd_i. We also keep the cubes filled on the way.

1. *Initial square.* $\Xi^{m+1}(f)$ fills the box over $[m]\times[1]_t$, line v, with lid $H^{m+1}(f)$ at $v=0$, end data $\psi\varphi(u)(f)$ at $t=0$ and $R_f(v)$ at $t=1$, and sides $\Xi^{m+1}(fd_i)$. Its end at $v=1$ is $G(f)$, which goes from $\psi\varphi(u)(f)$ to $R_f(1) = U_f$, with faces $G(fd_i)$.
2. *Successor cube.* $\Xi^{N+1}(f)$ is the end at $s=0$ of the filler of the box over $[m]\times[1]_t\times[1]_v$, line s, with lid $\Xi^N(f)$ at $s=1$, sides the cubes at fd_i, and end data

$$\begin{array}{ll} t=0: & \psi\varphi(u)(f), \\ t=1: & Q_f(s,v), \\ v=0: & \Theta_N(t,s), \\ v=1: & G(t), \text{ independent of } s. \end{array}$$

The pieces agree at the corners: at $t=1, v=0$ both are $R_f(s)$; at $t=1, v=1$ both are U_f; at $t=0$ all are $\psi\varphi(u)(f)$. At $s=1$ they are the edges of $\Xi^N(f)$: $Q_f(1,v) = R_f(v)$, $\Theta_N(t,1) = H^N(t)$. At $s=0$ they give the required edges of $\Xi^{N+1}(f)$: $Q_f(0,v) = R_f(v)$ and $\Theta_N(t,0) = H^{N+1}(t)$. On a face the end data are those of the face, by Lemma 4.3 and the face relations of ψ, H, Θ and G. The sides agree with the lid because the cube at fd_i starts at $\Xi^N(fd_i)$, which is $\Xi^N(f)$ on that face. The sides agree with each other where two faces meet, by the face relations one dimension lower. All boxes are filled in C itself: R_f and Q_f come from the full element u, so no truncation bound appears.

The prisms $G(f)$ are face compatible, so the values $G(f)(\mathrm{id}, h)$ for $(f,h) : [n] \to X \times [1]$ form an element of $\mathsf{D}A(X\times[1])$, natural along the strict maps, from $\psi\varphi(u)$ to u.

4.5 Naturality

The data at f are computed from the readings of the input along f and from the data at the faces of f. For $h : Y \to X$, the problem at f for the restriction of the input along h is the problem at hf for the input. The fillers are chosen uniformly, so equal problems have equal fillers, by induction on the dimension and on the stage. For $\psi\varphi \simeq \mathrm{id}$ one needs in addition that the readings R_f and Q_f of the restriction h^*u are those of u at hf, and that the restriction of $\varphi(u)$ along h is $\varphi(h^*u)$. So ψ and both homotopies are natural in X. A natural family of such cylinders is a section of the path type, by the definition of paths in Section 1. This proves Theorem 4.1.

Remark 4.4. No choice principle is used. The connectedness of the maps of the tower is witnessed by box fillers of C and of the C_N, chosen uniformly along the face maps. The internal argument from towers of connected maps would need dependent choice.

A Correspondence with the formalisation

Layer L9 of `core`, over the poset instance.

statement	files
$\mathsf{D}_N A = \mathsf{D}(K_N A)$, fibrancy	CCosk, CCoskHorn, CLocDN, CLocDNC
Proposition 2.1	CLocDN (Lim)
L, φ	CPostLimit (Product, Tower, Cone, Post); L fibrant: CPostFib
boxes	CBoxFill; over $[m] \times [1]$: CPostSq; over $[m] \times Y$: CPostBox; the boundary of the square and its four pieces: CPostCube (Geo, E4, E4Cmp)
readings, Lemma 4.3	CPostMap (DR), CPostK; for $u \in \mathsf{D}A$: CPostRead (β, Rd, Sq: R, $R2$, Q and its edges, Face)
ψ	CPostRec, CPostStep0/1/3, CPostInput
$\varphi\psi \simeq \mathrm{id}$	CPostStep2, CPostLine, CPostSquare, CPostAssemble; naturality CPostExt, CPostSqNat, CPostNat, CPostHomNat; the path: CPostConvergence (RightInverse.homotopy)
$\psi\varphi \simeq \mathrm{id}$	CPostInvIn (the recursion on $\varphi(u)$), CPostInv (G, Ξ, the cubes), CPostInvAsm (the cylinder, its ends), CPostInvNat (naturality; LeftInverse.homotopy)

Proposition 3.1 is not formalised. The fibrancy of L is CPostFib (ProductFib: countable products; HTFib: the path types of the edges; TowerFib, PostFib: L).